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§5 exploring $\mathbb{Z}$

* Chp. 4 → gave us the tools to do our usual algebra $-3 = [0,3] = [2,5]$

big hits

* Strong "350" Induction
* Euclid's Algorithm for GCD (greatest common denominator)
* The Fundamental Thm. of Arithmetic (# theory)
  - Euclid's Lemma
* The Infinitive of the Primes

Def. 1 - absolute value

Define for $a \in \mathbb{Z}$

$$|a| = \begin{cases} a & \text{if } a > 0 \\ -a & \text{if } a < 0 \end{cases}$$

$\boxed{\text{Thm. 8}}$ strong induction

Let $n \in \mathbb{Z}$ and let $P(n)$ be a statement

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Thm. 8 Strong Induction

Let $b \in \mathbb{Z}$ and let P be a statement.

Write $P(x)$ to indicate the statement holds for an integer x.

(i) If $P(b)$ holds base case!

(ii) $P(x)$ holding for all $b \leq x \leq n$ implies $P(n+1)$ holds induction hyp.

induction step

Then $P(x)$ holds for all $x \in \mathbb{Z}$ st. $x \geq b$.

ex) $P(x)$ the sum of the first x numbers is $\frac{x(x+1)}{2}$.

pf suppose (i) & (ii) hold but thm. doesn't hold

BWOC, let E be the set of $\mathbb{Z} \geq b$ for which the Thm. statement is not true.

First note every element of E is $> b$, since $P(b)$ holds. Thus b is a lower bound for E.

By Thm. 55, E has a minimum. well ordering principal

not lazy:

If $b \geq 0$, then $E \subseteq \mathbb{N}$, so WOP

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By Thm. 55, E has a minimum.

well ordering principal

not lazy:

If $b \geq 0$, then $E \subseteq \mathbb{N}$, so WOP says E has a least element.

If $b < 0$, take $E^{shift} = \{|b| + e : e \in E\}$.

So $E^{shift} \subseteq \mathbb{N}$ has a min m, so $m - |b|$ is a minimum for E, m.

Thus $P(x)$ holds for all $b \leq x \leq m-1$.

By (ii) $P(m-1+1)$ holds. Thus $P(m)$ holds.

Thus $m \notin E$! $\Rightarrow\Leftarrow$

we showed $m \in E$ which is the set of stuff that doesn't work, is in the set that does work, so there is a contradiction!

ex)

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ex)

12¢ or greater can be made with 5¢ & 4¢ stamps.

4¢ + 4¢ + 4¢ = 12¢

with n¢ & I know I can make (n-4)¢

add 4¢ to make n¢ ???

implies p(n+4-4) = n¢

Base case:

4¢+4¢+4¢ = 12¢

P(12) = 4+4+4, thus P(12) holds.

Suppose I can make n-3¢.

Add 4¢ stamp, now I have n+1¢.

If we want to make 13¢ from 12¢, I actually need 9¢.

14¢ need 10¢

15¢ need 11¢

16¢ need 12¢

Base case needs to include 12¢, 13¢, 14¢ & 15¢.

can't make 11¢ :(

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Suppose I can make n-3¢.

Add 4¢ stamp, now I have n+1 ¢.

If we want to make 13¢ from 12¢, I actually need a¢.

14¢ need 10¢

15¢ need 11¢

16¢ need 12¢

Base case needs to include 12¢, 13¢, 14¢ & 15¢.

Can't make 11¢ :(

~~P(15) holds b/c we take P(11) and add 4¢.~~

not true in this case!

↳ 15¢ = 3·5¢

pf

We proceed by strong induction.

For base cases 12 = 3·4¢, 13 = 5¢ + 2(4¢),

14 = 2(5¢) + 4¢, 15 = 3·5¢.

need 4 cases to show how we go back 4

For our induction step, suppose we have P(x) for all x st. 15 ≤ x ≤ n. WTS P(n+1) holds.

12

Since n+1 = n-3+4, we wish to consider P(n-3).

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For base cases $12 = 3 \cdot 4¢$, $13 = 5¢ + 2(4¢)$,

$14 = 2(5¢) + 4¢$, $15 = 3 \cdot 5¢$.

need 4 cases to show how we go back 4

For our induction step, suppose we have $P(x)$ for all x st. $15 \le x \le n$. WTS $P(n+1)$ holds.

12

Since $n+1 = n-3+4$, we wish to consider $P(n-3)$.

As $n \ge 15$, $n-3 \ge 12$.

$n-3 \ge 9$

Have to go to 15¢ to show it can be built by 5¢ & 4¢ stamps!

Thus $P(n-3)$ holds & so we use a 4¢ stamp to get $P(n+1)$, strong induction yields the result.

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Equivalence

Def 2:
If $\Delta(m,n) = \Delta(m',n')$, we say (m,n) & (m',n') are net difference equivalent.
In this case we write $(m,n) \sim (m',n')$

idea:

If I have \$50 & owe \$100, my net dif $= -50$

$$\Delta(50,100) = (0,50)$$

If I have \$1 & owe \$51, net dif $= -50$

$$\Delta(1,51) = (0,50)$$

$\Rightarrow$ These are the same if we're talking about how much I owe.

To formalize this, I define the equivalence class:

$$[0,50] = \{(m,n) \in \mathbb{N} : \Delta(m,n) = (0,50)\}$$

*****

Generally define

$$[m,n] = \{(a,b) \in \mathbb{N} \times \mathbb{N} : \Delta(a,b) = \Delta(m,n)\}$$

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Generally define

$$[m,n] = \{(a,b) \in \mathbb{N}\times\mathbb{N} : \Delta(a,b) = \Delta(m,n)\}$$

representatives

$$[0,50] = [30,80],\ [5,55],\ [10,6]$$

different reps., same result

equality of sets

$$\Delta(0,50) = (0,50) \neq (5,55)$$

$$\hookrightarrow = \Delta(5,55)$$

must take Δ of both

$(m,n) \sim (a,b)$ if $\Delta(m,n) = \Delta(a,b)$

$$[30,80] = \{(a,b) \in \mathbb{N}\times\mathbb{N} : (a,b) \sim (30,80)\}$$

The Integers are:

$$\mathbb{Z} = \{a : a = [m,n] \text{ for some } (m,n) \in \mathbb{N}\times\mathbb{N}\}$$

- basically the set of all equivalence classes

Def. $[m,n] + [r,s] = [m+r, n+s]$

If $[m,n] = [m',n']$ & $[r,s] = [r',s']$,

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The Integers are:

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- basically the set of all equivalence classes

Def. $[m,n] + [r,s] = [m+r, n+s]$

If $[m,n] = [m',n']$ & $[r,s] = [r',s']$,

then $[m,n] + [r,s] = [m'+r', n'+s']$

<u>ex</u>

$[100,0] = [200,100]$

$[0,12] = [32,44]$ ← this is $\cdot 12$

$[100,0] + [0,12] = [200,100] + [32,44]$

$[100,12] \overset{?}{=} [232,144]$

$100-12 \qquad 232-144$

$(88,0) = (88,0)$

$\boxed{\Delta[100,12] = \Delta[232,144] = (88,0).}$